4.1.1. Pandas - series creation and manipulation 16:15

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the `groupby` and `mean()` operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

- The user should enter a list of numbers separated by space when prompted.

Output Format:

- The program should display the mean of even and odd numbers separately.
- Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

Sample Test Cases +

Explorer seriesMa... Submit Debugger

```
import pandas as pd

# Take inputs from the user to create a list of numbers
numbers = list(map(int, input().split()))
```

Average time	Maximum time
0.015 s	0.028 s
14.83 ms	28.00 ms

3 out of 3 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1 28 ms Debug

Expected output	Actual output
1 2 3 4 5 6 7 8 9 10	1 2 3 4 5 6 7 8 9 10
Mean of even and odd numbers:	Mean of even and odd numbers:
Odd 5.0	Odd 5.0
Even 6.0	Even 6.0
dtype: float64	dtype: float64

Terminal Test cases

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4.1.2. Dictionary to dataframe 41:45

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row.

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row.

Delete a row:

- Take the row index to be deleted from the user.

Explorer datafram... Submit Debugger

```
import pandas as pd

# Provided dictionary of lists
data = {
```

Average time	Maximum time
0.115 s	0.120 s
115.00 ms	120.00 ms

1 out of 1 shown test case(s) passed

1 out of 1 hidden test case(s) passed

Test case 1 120 ms Debug

Expected output	Actual output
Original DataFrame:	Original DataFrame:
Name Age	Name Age
0 Alice 25	0 Alice 25
1 Bob 30	1 Bob 30
2 Charlie 35	2 Charlie 35
New name: Susan	New name: Susan

Terminal Test cases

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4.1.3. Student Information 26:17

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

- Display the first five rows of the data frame.
- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade is **'B'**).

Note:
Refer to the displayed test cases for better understanding.

Sample Test Cases +

studentin... Submit

```
import pandas as pd

# Read the text file into a DataFrame
file = input()
```

Average time 0.060 s 60.50 ms | Maximum time 0.078 s 78.00 ms

1 out of 1 shown test case(s) passed

1 out of 1 hidden test case(s) passed

Test case 1 78 ms Debug

Expected output	Actual output
studentdata.txt	studentdata.txt
First five rows:	First five rows:
Name Age Grade	Name Age Grade
0 John 25 A	0 John 25 A
1 Alin 22 B	1 Alin 22 B
2 Emma 24 A	2 Emma 24 A

Terminal Test cases

Prev Reset Submit Next

4.2.1. Month with the Highest Total Sales

15:40

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: `Date`, `Product`, `Quantity`, `Price`, and `City`.
- Group the data by Month and calculate the total sales for each month.
- Find the month with the highest total sales and display it.
- Also, display the total sales for the best month.

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Explorer | monthFor... | Submit | Debugger

```python
import pandas as pd

# Prompt the user for the file name
file_name = input()
```

Average time	Maximum time
0.033 s	0.067 s
33.33 ms	67.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 67 ms Debug

Expected output	Actual output
sales_data.csv	sales_data.csv
Best month: 2025-01	Best month: 2025-01
Total sales: $1210.00	Total sales: $1210.00

Terminal | Test cases

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4.2.2. Best Selling Product

16:57

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: `Date`, `Product`, `Quantity`, `Price`, and `City`.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Explorer | monthFor... | Submit | Debugger

```python
import pandas as pd

# Prompt the user for the file name
file_name = input()
```

Average time 0.022 s 22.00 ms | Maximum time 0.042 s 42.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 42 ms Debug

Expected output	Actual output
sales_data.csv	sales_data.csv
Best selling product: Product A	Best selling product: Product A
Total quantity sold: 23	Total quantity sold: 23

Terminal | Test cases

< Prev | Reset | Submit | Next >

4.2.3. City that Sold the Most Products 05:00

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: `Date`, `Product`, `Quantity`, `Price`, and `City`.
- Group the data by `City` and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Explorer monthFor... Submit Debugger

```
import pandas as pd

# Prompt the user for the file name
file_name = input()
```

Average time	Maximum time
0.030 s	0.070 s
30.00 ms	70.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 70 ms Debug

Expected output	Actual output
sales_data.csv	sales_data.csv
City sold the most products: Los Angeles	City sold the most products: Los Angeles

Terminal Test cases

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4.2.4. Most Frequently Sold Product Pairs

34:15

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the following columns: Date, Product, Quantity, Price, and City.
- For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
- Output the product pair/s that was sold most frequently.

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Explorer frequentl... Submit Debugger

```python
import pandas as pd
from itertools import combinations
from collections import Counter

```

Average time	Maximum time
0.024 s	0.051 s
24.33 ms	51.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 51 ms Debug

Expected output	Actual output
sales_data.csv	sales_data.csv
Product A and Product B: 2 times	Product A and Product B: 2 times
Product A and Product C: 2 times	Product A and Product C: 2 times

Terminal Test cases

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4.2.5. Titanic Dataset Analysis and Data Cleaning 08:59

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

1. Display the first 5 rows of the dataset.
2. Display the last 5 rows of the dataset.
3. Get the shape of the dataset (number of rows and columns).
4. Get a summary of the dataset (using `.info()`).
5. Get basic statistics (mean, standard deviation, etc.) of the dataset using `.describe()`.
6. Check for missing values and display the count of missing values for each column.
7. Fill missing values in the 'Age' column with the median age.
8. Fill missing values in the 'Embarked' column with the most frequent value (`mode()`).
9. Drop the 'Cabin' column due to many missing values.
10. Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

Explorer titanicDat... Submit Debugger

```python
import pandas as pd
import numpy as np

# Load the Titanic dataset
```

Average time	Maximum time
0.453 s 453.00 ms	0.453 s 453.00 ms

1 out of 1 shown test case(s) passed

Test case 1 453 ms Debug

Expected output	Actual output
PassengerId Survived Pclass ... Fare Cabin Embarked	PassengerId Survived Pclass ... Fare Cabin Embarked
0 1 0 3 ... 7.2500 NaN S	0 1 0 3 ... 7.2500 NaN S
1 2 1 1 ... 71.2833 C85 C	1 2 1 1 ... 71.2833 C85 C
2 3 1 3 ... 7.9250 NaN S	2 3 1 3 ... 7.9250 NaN S

Terminal Test cases

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4.2.6. Titanic Dataset Analysis and Data Cleaning - 2

29:09

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.
2. Convert the 'Sex' column to numeric values (male: 0, female: 1).
3. One-hot encode the 'Embarked' column, dropping the first category.
4. Get the mean age of passengers.
5. Get the median fare of passengers.
6. Get the number of passengers by class.
7. Get the number of passengers by gender.
8. Get the number of passengers by survival status.
9. Calculate the survival rate of passengers.
10. Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below.

P as	S ur	Pc	N			Si	P	Ti		C	E m

Explorer

titanicDat...

Submit

```
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
data['FamilySize'] = data['SibSp'] + data['Parch']

# # 1. Create a new column ‘IsAlone’ (1 if alone, 0 otherwise)
# data['Alone']=np.where(data["FamilySize"]==0,1,0)

# # 2. Convert ‘Sex’ to numeric (male: 0, female: 1)
# data['Sex']=data['Sex'].map({'male':0,'female':1})

# # 3. One-hot encode the ‘Embarked’ column
# data=pd.get_dummies(data,columns=['Embarked'],drop_first=True)

# # 4. Get the mean age of passengers
```

Debugger

Terminal Test cases

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4.2.7. Titanic Dataset Analysis and Data Cleaning - 3

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Calculate the survival rate by class.
2. Calculate the survival rate by embarkation location (Embarked_S).
3. Calculate the survival rate by family size (FamilySize).
4. Calculate the survival rate by being alone (IsAlone).
5. Get the average fare by passenger class (Pclass).
6. Get the average age by passenger class (Pclass).
7. Get the average age by survival status (Survived).
8. Get the average fare by survival status (Survived).
9. Get the number of survivors by class (Pclass).
10. Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below,

P as	S ur	Pc	N			Si	P	Ti		C	E m

titanicDat...

```
import pandas as pd
import numpy as np

# Load the Titanic dataset
```

Average time	Maximum time
0.139 s	0.139 s
139.00 ms	139.00 ms

1 out of 1 shown test case(s) passed

Test case 1 139 ms Debug

Expected output	Actual output
Pclass	Pclass
1 0.629630	1 0.629630
2 0.472826	2 0.472826
3 0.242363	3 0.242363
Name: Survived, dtype: float64	Name: Survived, dtype: float64
Embarked_S	Embarked_S

Terminal Test cases

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4.2.8. Titanic Dataset Analysis and Data Cleaning - 4

26:13

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Get the number of survivors by gender (Sex).
2. Get the number of non-survivors by gender (Sex).
3. Get the number of survivors by embarkation location (Embarked_S).
4. Get the number of non-survivors by embarkation location (Embarked_S).
5. Calculate the percentage of children (Age < 18) who survived.
6. Calculate the percentage of adults (Age >= 18) who survived.
7. Get the median age of survivors.
8. Get the median age of non-survivors.
9. Get the median fare of survivors.
10. Get the median fare of non-survivors.

The Titanic dataset contains columns as shown below,

P as	S ur	Pc	N			Si	P	Ti		C	E m

titanicDat... Submit

```python
import pandas as pd
import numpy as np

# Load the Titanic dataset
```

Average time 0.108 s 108.00 ms

Maximum time 0.108 s 108.00 ms

1 out of 1 shown test case(s) passed

Test case 1 108 ms Debug

Expected output	Actual output
female 233	female 233
male 109	male 109
Name: Sex, dtype: int64	Name: Sex, dtype: int64
male 468	male 468
female 81	female 81
Name: Sex, dtype: int64	Name: Sex, dtype: int64

Terminal Test cases